

Receiver-Guided Reinforcement Learning-based Congestion Control for IoT

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Abstract—The abstract goes here.

Index Terms—IoT, TCP, congestion control, reinforcement learning.

I. INTRODUCTION

THIS part is to be completed.

II. RELATED WORKS

III. SYSTEM DESIGN

Our proposal contains three modules in the receiver side for adjusting sender *cwnd*, Receiver Measurement Module, Window Decision Module, and Window Enforcement Module. Additionally, TCP sockets in both sender and receiver are extended or modified for adding interfaces to achieved our features.

A. Message types definition

The receiver TCP socket is extended to support communication with the RL agent. Three types of messages are defined:

- **State**: carries the observed network state and is sent to the agent for storage in the replay buffer.
- **Action Request**: triggers the agent to infer and return a control action based on the current state.
- **Connection Close**: notifies the agent that a flow has terminated, allowing it to clear the corresponding internal states.

B. Receiver Measurement Module

Our scheme relies on reliable state observations to ensure stable training. Therefore, the states in our design are required to reflect the congestion extent while remaining observable at the receiver. Despite RTT is a fine-grained and valuable indicator of congestion, it is difficult to directly observe in the receiver side.

To address this limitation, we derive a receiver-side congestion extent metric based on the *cwnd* elapsed time. We extend the sender TCP feature to include its *cwnd* in TCP header. In standard TCP congestion avoidance, the congestion window *cwnd* is increased by 1 MSS after an entire *cwnd* bytes has been acknowledged, resulting in an approximate growth rate

of 1 MSS per RTT, in which the method is named Additive Increment.

We utilise Additive Increment growing feature to enable the receiver to estimate network congestion extent by calculating the elapsed time for a flow to complete its *cwnd*. In particular, upon every packet receive event that includes a *cwnd* value different from the last value received, the Receiver Measurement Module establishes a new estimation task. A new task is initialised with the current observation (*o*) and a threshold (τ), e.g., for flow *i* started at time t_0 and estimated at time t_j , defined as below:

$$o_{i,t_j} \triangleq [B_i(t_j), L_i(t_j), t_j] \quad (1)$$

$$B_i(t_j) = \sum_{n=t_0}^{t_j} RxBytes_{i,n}$$

$$L_i(t_j) = \sum_{n=t_0}^{t_j} Losses_{i,n}$$

$$\tau_{i,t_j} = Seq_{i,t_j} + cwnd_{i,t_j} \quad (2)$$

An estimation task is completed at $t_{j+\Delta}$ when a coming packet is seen with a sequence number satisfying $Seq_{i,t_{j+\Delta}} > \tau_{i,t_j}$. We calculate this task duration as the *cwnd* elapsed time i.e., $t_{elapsed} = t_{j+\Delta} - t_j$. The $t_{elapsed}$ is then used to calculate the goodput (g_{i,t_j}) and the packet loss rate (ℓ_{i,t_j}) during the estimation task, where $g_{i,t_j} = \frac{B_i(t_{j+\Delta}) - B_i(t_j)}{t_{elapsed}}$ and $\ell_{i,t_j} = \frac{L_i(t_{j+\Delta}) - L_i(t_j)}{t_{elapsed}}$. We then form a state message according to $s_{i,t_j} = S(o_{i,t_{j+\Delta}}, o_{i,t_j}; \tau_{i,t_j})$, where the generation process is defined as follows:

$$S(o_{i,t_{j+\Delta}}, o_{i,t_j}; \tau_{i,t_j}) = [g_{i,t_j}, \ell_{i,t_j}, t_{elapsed}]$$

The Receiver Measurement and State Generation process can be seen in Algorithm 1.

C. Window Decision Module

Window Decision Module consists of a Reinforcement Learning model to output a scaling factor to underlying TCP socket. An enforced congestion window is calculated with the factor by the underlying TCP socket and included in ACK to the sender. The sender will overwrite its *cwnd* by the enforced congestion window. Therefore, to implement a continuous control to modified TCP, we choose Deep Deterministic Policy Gradient (DDPG) as agent in our design

Our agent consists of an actor network determining a deterministic action $a = \mu(s; \theta^\mu)$ parameterised by θ^μ and

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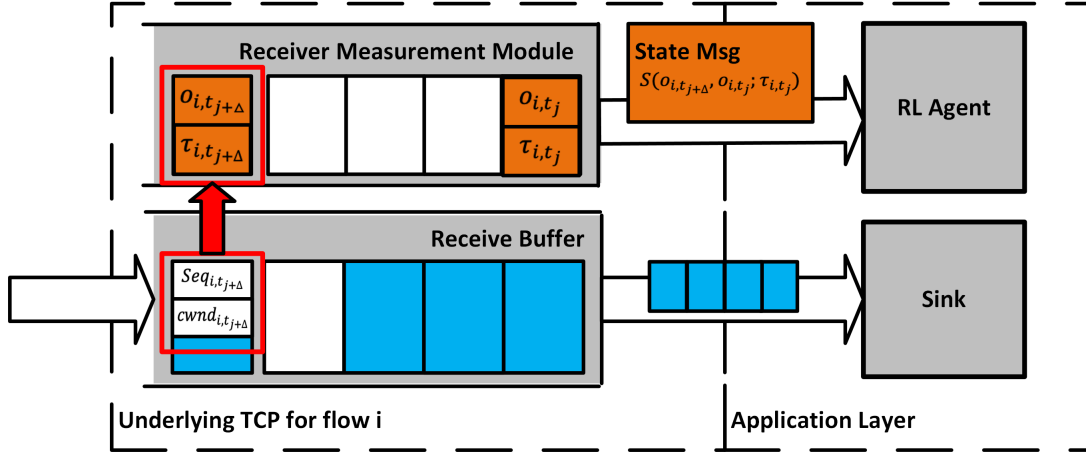


Fig. 1. An example of estimation task completion of flow i .

Algorithm 1 Receiver Measurement and State Generation

Input: Flow Id i , Time t_j , Arrived Packet p_{i,t_j} , Flow Observation Queue q_i

Output: State message s_{i,t_j}

- 1: Extract Seq_{i,t_j} , $cwnd_{i,t_j}$, and $RxBytes_{i,t_j}$ from p_{i,t_j}
 - 2: $\tau_{i,t_j} = Seq_{i,t_j} + cwnd_{i,t_j}$
 - 3: Obtain $Losses_{i,t_j}$ from $TcpSocket_i$
 - 4: $B_i(t_j) = B_i(t_{j-1}) + RxBytes_{i,t_j}$
 - 5: $L_i(t_j) = L_i(t_{j-1}) + Losses_{i,t_j}$
 - 6: $o_{i,t_j} = [B_i(t_j), L_i(t_j), t_j]$
 - 7: Obtain o_i^{head} and τ_i^{head} from the head of q_i
 - 8: **if** $Seq_{i,t_j} \geq \tau_i^{head}$ **then**
 - 9: Dequeue o_i^{head} and τ_i^{head} from q_i
 - 10: $s_{i,t_j} = S(o_{i,t_j}, o_i^{head})$
 - 11: Send s_{i,t_j} as State Message to Agent
 - 12: **else if** $cwnd_{i,t_j} \neq cwnd_{i,t_{j-1}}$ **then**
 - 13: Enqueue o_{i,t_j} and τ_{i,t_j} into q_i
 - 14: **end if**
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D. Window Enforcement Module

The sender is modified to leverage the window decision as a factor to update its $cwnd$ as following:

$$cwnd \leftarrow cwnd \times Scaling_Factor^{Decision}$$

E. Enforcing Window Generation by RL Agent

IV. SIMULATION SETUP AND RESULTS

V. CONCLUSION

The conclusion goes here.

ACKNOWLEDGMENT

The authors would like to thank...

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a critic network to approximate the expected future return, defined as $Q(s, a; \theta^Q) = \mathbb{E}[R_t | s_t, a_t]$ with parameters θ^Q . The return R_t is the sum of discounted future return, i.e., $R_t = \sum_{i=t}^T \gamma^{i-t} r(s_i, a_i)$. The critic network is updated by minimising the mean square error between its output and a target function (y_t), shown as follows:

$$L(\theta^Q) = \mathbb{E}[(Q(s_t, a_t; \theta^Q) - y_t)^2]$$

where $y_t = r(s_t, a_t) + \gamma Q(s_{t+1}, \mu(s_{t+1}; \theta^\mu); \theta^Q)$. Subsequently, with the updated critic network, policy performance of current actor network is predicted, expressed as $J(\theta^\mu) = \mathbb{E}[Q(s_t, \mu(s_t; \theta^\mu))]$. Then the actor network is updated by calculating the gradient of the policy performance, which, by applying the chain rule, is computed as:

$$\nabla_{\theta^\mu} J(\theta^\mu) = \mathbb{E}[\nabla_{a_t} Q(s_t, \mu(s_t; \theta^\mu)) \nabla_{\theta^\mu} \mu(s_t; \theta^\mu)]$$